

Intermediate Level Report

SuperStore Sales Data Cleaning & Business Analysis

1. Introduction

This project focuses on cleaning, transforming, and analyzing a **SuperStore sales dataset** using **Python (Pandas)** and **Microsoft Power BI**. The objective of the Intermediate-level task is to move beyond basic visualization and perform meaningful business analysis involving sales, profit, customers, products, regions, segments, and returns.

The original dataset contains **5,901 records and 23 columns**. It includes information such as Order ID, Order Date, Ship Date, Customer, Segment, Region, Product, Category, Sales, Quantity, Profit, Returns, and Payment Mode.

The complete workflow followed in this project is:

Raw Dataset → Data Cleaning → Feature Engineering → Power BI Analysis → Dashboard → Business Insights & Recommendations

2. Dataset Description

The dataset contains transaction-level information about SuperStore orders.

Important fields include:

Column	Description
Order ID	Unique identifier of an order
Order Date	Date on which the order was placed
Ship Date	Date on which the order was shipped
Ship Mode	Shipping method used
Customer ID	Customer identifier
Customer Name	Customer name
Segment	Customer segment
City	Customer city
State	Customer state
Region	Business region
Category	Product category
Sub-Category	Product sub-category
Product Name	Product description
Sales	Revenue generated
Quantity	Number of items ordered

Column	Description
Profit	Profit generated
Returns	Return information
Payment Mode	Payment method

The dataset provides enough information to perform customer, product, geographical, profitability, and return analysis.

3. Data Cleaning Process

The dataset was first loaded into **Python using Pandas** and inspected using functions such as `info()`, `head()`, `describe()`, and missing-value analysis.

3.1 Column Name Cleaning

The original dataset contained an incorrectly formatted column name:

Row ID+O6G3A1:R6

It was renamed to:

Row ID

This makes the dataset easier to understand and work with.

3.2 Removing Empty Columns

Two columns were found to contain no values:

- ind1
- ind2

Since these columns did not provide any useful information, they were removed.

3.3 Duplicate Check

The dataset was checked for completely duplicated records. Duplicate rows were removed where applicable.

However, repeated **Order IDs were not removed**, because a single order can contain multiple products and therefore legitimately appear in multiple records.

3.4 Date Conversion

Order Date and Ship Date were originally stored as text/object values.

They were converted into proper datetime format using Pandas.

This allowed further analysis such as:

- Monthly sales
- Yearly sales
- Shipping duration
- Sales trends

3.5 Return Data Cleaning

The Returns column contained values for returned orders and missing values for the remaining records.

Instead of deleting the missing records, a new column called **Return Status** was created:

- Returned
- Not Returned

This preserves all transaction records while making return analysis easier.

3.6 Text Cleaning

Text fields were stripped of unnecessary leading and trailing spaces to maintain consistency in categorical analysis.

3.7 Data Validation

Numerical fields such as:

- Sales
- Quantity
- Profit

were checked for unusual or invalid values.

Negative profit values were **not removed**, because a negative profit represents an actual business loss and is useful for profitability analysis.

4. Feature Engineering

Additional columns were created to make the dataset more useful for Power BI analysis.

Shipping Days

Calculated using:

Shipping Days = Ship Date – Order Date

This allows shipping performance to be analyzed.

Year

Extracted from Order Date.

Month

Extracted from Order Date for monthly trend analysis.

Month Number

Created to maintain the correct chronological order of months.

Month Year

Created for monthly Power BI trend visualizations.

Profit Margin %

Calculated as:

Profit Margin % = Profit ÷ Sales × 100

This provides a better understanding of profitability rather than looking only at sales.

5. Power BI Analysis

After cleaning, the processed dataset was imported into **Microsoft Power BI**.

The dashboard was designed to provide an interactive view of business performance.

Main Visualizations

5.1 Sales and Profit by Category

A clustered column chart was used to compare sales and profit across:

- Technology
- Office Supplies
- Furniture

This helps identify categories that generate high revenue and determine whether that revenue is accompanied by strong profitability.

5.2 Sales by Region

A bar chart was created to compare sales across:

- West
- East
- Central
- South

This helps identify the strongest-performing geographical regions.

5.3 Sales and Profit by Customer Segment

A clustered column chart compares:

- Consumer
- Corporate
- Home Office

The visualization helps determine which customer segment contributes most to revenue and profit.

5.4 Sales by Sub-Category

A horizontal bar chart was used to compare sales across product sub-categories.

This can identify high-demand areas such as Phones, Chairs, Binders, Storage, and Accessories.

5.5 Monthly Sales and Profit

A trend visualization was created using the month field.

It helps identify:

- Increasing or decreasing sales
- Seasonal patterns
- Changes in profitability

5.6 Top Products

A Top 10 product analysis was created based on sales.

This identifies products that contribute significantly to overall revenue.

5.7 Top Customers

A Top 10 customer analysis was created based on sales to identify the customers contributing the most revenue.

5.8 Return Analysis

A donut chart was created using **Return Status** to compare returned and non-returned transactions.

Another visualization compares return counts across product categories.

6. Business Insights

The dashboard allows several important business questions to be answered.

Sales Performance

The analysis shows how sales are distributed across different categories, regions, products, and customer segments.

Profitability

Sales alone do not represent business performance. By comparing Sales with Profit, the company can identify areas where high revenue does not necessarily produce high profit.

Customer Analysis

The customer and segment analysis helps identify the most valuable customer groups and individual customers.

Product Analysis

Top-product and sub-category analysis helps identify products with strong demand and potential opportunities for increasing revenue.

Regional Analysis

Comparing regions helps identify geographical areas that are performing strongly and areas requiring further investigation.

Return Analysis

Return analysis can help determine which categories have a higher number of returned orders and may require investigation into product quality, customer expectations, or order fulfillment.

7. Recommendations

Based on the analysis, the following business actions can be considered:

1. **Focus on profitable products**, rather than evaluating products only by sales volume.

2. **Investigate low-profit or loss-making products** to understand pricing, discounting, and cost issues.
 3. **Strengthen high-performing customer segments** through targeted offers and customer retention strategies.
 4. **Analyze weaker regions** and identify reasons for lower sales or profitability.
 5. **Monitor return-heavy categories** to identify potential product or fulfillment problems.
 6. **Use monthly trends** to improve inventory planning and sales campaigns during stronger periods.
 7. **Monitor high-value customers** and develop retention strategies for customers contributing significantly to revenue.
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8. Conclusion

The Intermediate-level project demonstrates a complete data-analysis workflow, beginning with raw data and ending with an interactive Power BI dashboard.

The project involved **data inspection, cleaning, validation, transformation, feature engineering, visualization, business analysis, and recommendation generation**.

Python and Pandas were used to prepare the dataset, while Power BI was used to transform the cleaned data into interactive business visualizations.

The major improvement over a Beginner-level dashboard is that the project does not only display numbers. It attempts to answer important business questions such as:

Where is revenue coming from?

Which areas are profitable?

Which customers and products are most valuable?

Where are returns occurring?

What areas require business attention?

Thus, the project demonstrates practical skills in **data cleaning, exploratory business analysis, visualization, and insight generation**.